PSP SYS Monolith Aurora PostgreSQL migration

Current database objects layout

Primary/Monolith database

<table>
<tr><th></th><th>OWNER</th><th>OBJECT_TYPE</th><th>OBJECT_COUNT</th><th>Notes</th></tr>
<tr><td>1</td><td>PSPADM</td><td>DATABASE LINK</td><td>2</td><td></td></tr>
<tr><td>2</td><td>PSPADM</td><td>FUNCTION</td><td>16</td><td></td></tr>
<tr><td>3</td><td>PSPADM</td><td>INDEX</td><td>796</td><td></td></tr>
<tr><td>4</td><td>PSPADM</td><td>JOB</td><td>1</td><td></td></tr>
<tr><td>5</td><td>PSPADM</td><td>LOB</td><td>25</td><td>Tables with LOB columns
<table>
<tr><th></th><th>TABLE_NAME</th><th>COLUMNS</th></tr>
<tr><td>1</td><td>AT_WIDGET</td><td>WIDGET_CONFIGURATION</td></tr>
<tr><td>2</td><td>PSP_ACHENROLLMENT_FILE</td><td>FILE_CONTENT, FILE_CONTENT_ENC</td></tr>
<tr><td>3</td><td>PSP_CHECK_PRINT_SIGNATURE</td><td>SIGNATURE</td></tr>
<tr><td>4</td><td>PSP_ENTITLEMENT_MESSAGE</td><td>MESSAGE, MESSAGE_ENC</td></tr>
<tr><td>5</td><td>PSP_ENTITY_UPDATE</td><td>CHANGED_ATTRIBUTES</td></tr>
<tr><td>6</td><td>PSP_LEDGER_OPERATION_JOB</td><td>ORIGINAL_FILE, PROCESSED_FILE</td></tr>
<tr><td>7</td><td>PSP_MESSAGE_LOG</td><td>REQUEST_LOG, RESPONSE_LOG</td></tr>
<tr><td>8</td><td>PSP_RTBAUTOMATIONBACKUP</td><td>RTB_BACKUP</td></tr>
<tr><td>9</td><td>PSP_SAVED_REPORTS</td><td>QUERY</td></tr>
<tr><td>10</td><td>PSP_SOURCESYS_PRINTEDCHK_INFO</td><td>BANK_LOGO, SOURCE_SYSTEM_LOGO</td></tr>
<tr><td>11</td><td>PSP_SQL_EXECUTION_LOG_ENTRY</td><td>S_Q_L</td></tr>
<tr><td>12</td><td>PSP_STATE_REPORT_OUTPUT</td><td>REPORT_OUTPUT</td></tr>
<tr><td>13</td><td>PSP_SUICREDITS_JOB</td><td>PROCESSED_FILE</td></tr>
<tr><td>14</td><td>PSP_TAX_CREDITS9061</td><td>FORM9061</td></tr>
<tr><td>15</td><td>PSP_TAX_CREDITS_APPLICATION</td><td>SIGNED_DOCUMENT,
UNSIGNED_DOCUMENT</td></tr>
<tr><td>16</td><td>QUEST_SL_TEMP_EXPLAIN2</td><td>OTHER_XML</td></tr>
<tr><td>17</td><td>SYS_EXPORT_SCHEMA_01</td><td>XML_CLOB</td></tr>
<tr><td>18</td><td>SYS_EXPORT_SCHEMA_02</td><td>XML_CLOB</td></tr>
<tr><td>19</td><td>TOAD_PLAN_TABLE</td><td>OTHER_XML</td></tr>
</table>
</td></tr>
<tr><td>6</td><td>PSPADM</td><td>PACKAGE</td><td>3</td><td></td></tr>
<tr><td>7</td><td>PSPADM</td><td>PACKAGE BODY</td><td>3</td><td></td></tr>
<tr><td>8</td><td>PSPADM</td><td>PROCEDURE</td><td>29</td><td></td></tr>
</table>

9	PSPADM	SEQUENCE	71	
10	PSPADM	TABLE	394	
11	PSPADM	TABLE PARTITION	347	
12	PSPADM	TRIGGER	24	
13	PSPADM	TYPE	1	
14	PSPADM	VIEW	4	

	TABLE_NAME	No. Of Partitions
1	PSP_BATCH_JOB_AUDIT_LOG	74
2	PSP_ENTRY_DETAIL_RECORD	38
3	PSP_FINANCIAL_TRANSACTION	19
4	PSP_FINANCIAL_TRANS_STATE	38
5	PSP_LEDGER_BALANCE	10
6	PSP_MONEY_MOVEMENT_TRANSACTION	20
7	PSP_PAYCHECK	10
8	PSP_PAYCHECK_SPLIT	10
9	TPR_METADATA	128

Comparison of constraints between source and target: https://docs.google.com/spreadsheets/d/16tksMItUI_8Jr0chnwSgD86SFWuM5WK6mTkuidzBldQ/edit#gid=225718943

Assessment report

Summary

Monolith

Figure: Conversion statistics for database storage objects

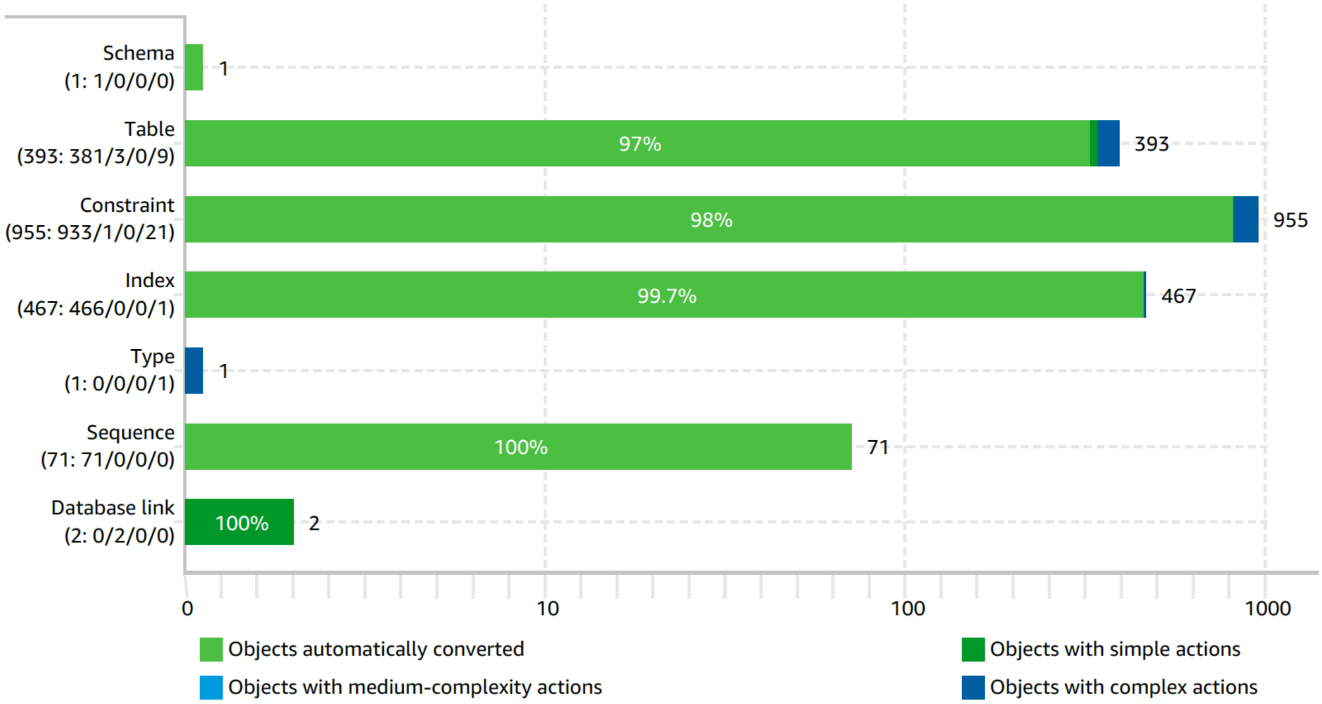
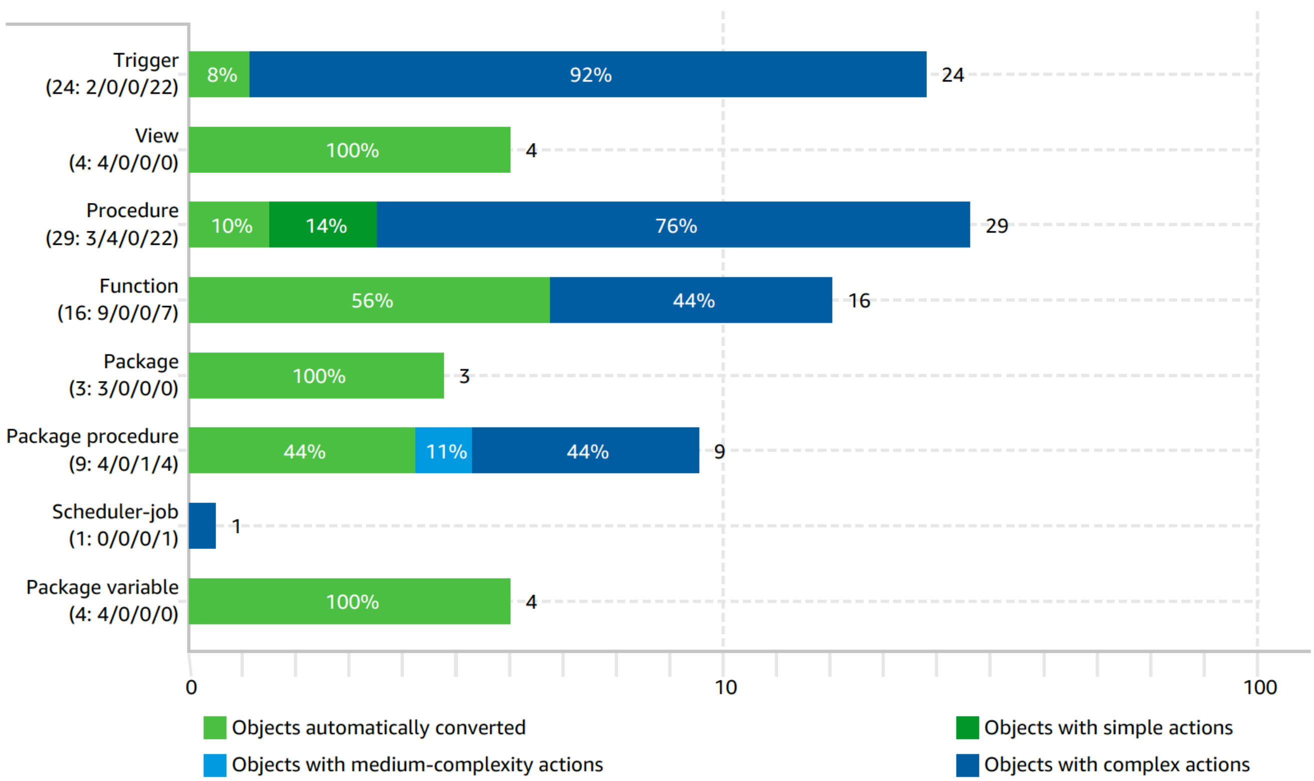


Figure: Conversion statistics for database code objects



No statistics available for database code objects for SYS env.

Full report

monolith: psp_sys_monolith_sct_assessment_rpt.pdf

Oracle to Aurora PostgreSQL

Limitations while using DMS (Oracle to Aurora PostgreSQL)

- Limitation on using Oracle RDS as source

https://docs.aws.amazon.com/dms/latest/userguide/CHAP_Source.Oracle.html#CHAP_Source.Oracle.Limitations

- Limitations on using PostgreSQL as a target for AWS Database Migration Service

https://docs.aws.amazon.com/dms/latest/userguide/CHAP_Target.PostgreSQL.html#CHAP_Target.PostgreSQL.Limitations

- Table partitioning limitations as of PostgreSQL 11 (reference: https://www.postgresql.org/docs/11/ddl-partitioning.html#DDL-PARTITIONING-DECLARATIVE)

The following limitations apply to partitioned tables:

- There is no way to create an exclusion constraint spanning all partitions; it is only possible to constrain each leaf partition individually.
- Unique constraints (and hence primary keys) on partitioned tables must include all the partition key columns. This limitation exists because PostgreSQL can only enforce uniqueness in each partition individually.
- While primary keys are supported on partitioned tables, foreign keys referencing partitioned tables are not supported. (Foreign key references from a partitioned table to some other table are supported.)
- `BEFORE ROW` triggers, if necessary, must be defined on individual partitions, not the partitioned table.
- Mixing temporary and permanent relations in the same partition tree is not allowed. Hence, if the partitioned table is permanent, so must be its partitions and likewise if the partitioned table is temporary. When using temporary relations, all members of the partition tree have to be from the same session.

Schema Conversion

Issue #1: DDLs for Check constraints do not get generated correctly.

Solution: We would need to generate it manually for APG and create check constraints accordingly on APG target database.

Issue #2: SCT do not generate DDL for FK referring to partitioned tables.

Solution: Manually compare and create missing FKs accordingly.

Issue #3: Table psp_financial_transaction table. settlement_date is coming as NULL in insert statement.

Issue #4: Table psp_event_detail_type migration getting failed with check constraint violation.

Database objects layout after migration using DMS

Initial layout

	schema_name	object_type	object_count
1	pspadm	PARTITIONED INDEX	56
2	pspadm	INDEX	2224
3	pspadm	PARTITIONED TABLE	9
4	pspadm	SEQUENCE	71
5	pspadm	TABLE	704
6	pspadm	TYPE	12
7	pspadm	VIEW	32

Code Issue

Functions

- pspadm.get_or_pending_agency_id
- pspadm.quest_sl_temp_explain1
- pspadm.quest_sl_temp_explain2
- pspadm.sys_temp_fbt_iud

```
1. FUNCTION pspadm.get_or_pending_agency_id
psql:CREATE-FUNCTION.sql:568: ERROR:  "mmt.money_movement_transaction_seq" is not a known variable
LINE 10:         INTO STRICT source_company_id, mmt.money_movement_tr...

2. FUNCTION pspadm.quest_sl_temp_explain1                                                ^
psql:CREATE-FUNCTION.sql:1373: ERROR:  syntax error at or near "timestamp"
LINE 2: ...t_id CHARACTER VARYING, plan_id DOUBLE PRECISION, timestamp ...
                                                             ^
3. FUNCTION pspadm.quest_sl_temp_explain2
psql:CREATE-FUNCTION.sql:1432: ERROR:  syntax error at or near "timestamp"
LINE 2: ...t_id CHARACTER VARYING, plan_id DOUBLE PRECISION, timestamp ...
                                                             ^
4. FUNCTION pspadm.sys_temp_fbt_iud
psql:CREATE-FUNCTION.sql:5281: ERROR:  syntax error at or near "#"
LINE 23: ...RT INTO sys_temp_fbt$tmp(schema,object_name,OBJECT#,rid,acti...
                                                              ^
```

Procedures

- pspadm.prc_offload$update_nacha_file_trace_number$959a4e53
- pspadm.prc_remove_company
- pspadm.prc_remove_company2
- pspadm.prc_remove_record_test

```
1. PROCEDURE pspadm.prc_offload$update_nacha_file_trace_number$959a4e53
psql:CREATE-PROCEDURE.sql:1506: ERROR:  "sql$rowcount" is not a known variable
LINE 63:     GET DIAGNOSTICS sql$rowcount = ROW_COUNT;
                             ^
2. PROCEDURE pspadm.prc_remove_company
psql:CREATE-PROCEDURE.sql:2563: ERROR:  type "pspadm.prc_remove_company$c1$r[]" does not exist
LINE 13:     UserConstraints pspadm.prc_remove_company$c1$r [];
                             ^
3. PROCEDURE pspadm.prc_remove_company2
psql:CREATE-PROCEDURE.sql:2722: ERROR:  type "pspadm.prc_remove_company2$c1$r[]" does not exist
LINE 13:     UserConstraints pspadm.prc_remove_company2$c1$r [];
                             ^
4. PROCEDURE pspadm.prc_remove_record_test
psql:CREATE-PROCEDURE.sql:11711: ERROR:  type "pspadm.prc_remove_record_test$c1$r[]" does not exist
LINE 13:     UserConstraints pspadm.prc_remove_record_test$c1$r [];
                             ^
```

Solution: Able to fix code of below w.r.t TYPE variables

- pspadm.prc_remove_company
- pspadm.prc_remove_company2
- pspadm.prc_remove_record_test

Triggers

Our apps connect to database as 'pspapp' user. Employee & Individual tables are using inheritance in hibernate layer. So, when there are bulk inserts or bulk updates on these tables, hibernate creates Global Temporary Tables for staging purposes. Temporary tables ht_psp_employee & ht_psp_individual are created on the fly in our workflow. These tables are created under 'pspadm' user and ‘pspapp' user does not have required privileges on 'ht_psp_employee' & 'ht_psp_individual' tables because these tables are created on the fly.

```
psql:CREATE-TRIGGER.sql:199: ERROR:  relation "pspadm.ht_psp_agency_check_batch" does not exist
psql:CREATE-TRIGGER.sql:207: ERROR:  relation "pspadm.ht_psp_agency_id_requirement" does not exist
psql:CREATE-TRIGGER.sql:215: ERROR:  relation "pspadm.ht_psp_bpcompany_service_info" does not exist
psql:CREATE-TRIGGER.sql:223: ERROR:  relation "pspadm.ht_psp_cdcompany_service_info" does not exist
psql:CREATE-TRIGGER.sql:231: ERROR:  relation "pspadm.ht_psp_check_print_batch" does not exist
psql:CREATE-TRIGGER.sql:239: ERROR:  relation "pspadm.ht_psp_company_paycheck_batch" does not exist
psql:CREATE-TRIGGER.sql:247: ERROR:  relation "pspadm.ht_psp_company_service" does not exist
psql:CREATE-TRIGGER.sql:255: ERROR:  relation "pspadm.ht_psp_contact" does not exist
psql:CREATE-TRIGGER.sql:263: ERROR:  relation "pspadm.ht_psp_ddcompany_service_info" does not exist
psql:CREATE-TRIGGER.sql:271: ERROR:  relation "pspadm.ht_psp_dep_freq_ledger_operati" does not exist
psql:CREATE-TRIGGER.sql:279: ERROR:  relation "pspadm.ht_psp_deposit_frequency_req" does not exist
psql:CREATE-TRIGGER.sql:287: ERROR:  relation "pspadm.ht_psp_edi_tax_file" does not exist
psql:CREATE-TRIGGER.sql:295: ERROR:  relation "pspadm.ht_psp_eftps_file" does not exist
psql:CREATE-TRIGGER.sql:303: ERROR:  relation "pspadm.ht_psp_employee" does not exist
psql:CREATE-TRIGGER.sql:311: ERROR:  relation "pspadm.ht_psp_individual" does not exist
psql:CREATE-TRIGGER.sql:319: ERROR:  relation "pspadm.ht_psp_ledger_operation" does not exist
psql:CREATE-TRIGGER.sql:327: ERROR:  relation "pspadm.ht_psp_manual_requirement" does not exist
psql:CREATE-TRIGGER.sql:335: ERROR:  relation "pspadm.ht_psp_payment_method_requirem" does not exist
psql:CREATE-TRIGGER.sql:343: ERROR:  relation "pspadm.ht_psp_payment_requirement" does not exist
psql:CREATE-TRIGGER.sql:351: ERROR:  relation "pspadm.ht_psp_racompany_service_info" does not exist
psql:CREATE-TRIGGER.sql:359: ERROR:  relation "pspadm.ht_psp_rate_ledger_operation" does not exist
psql:CREATE-TRIGGER.sql:367: ERROR:  relation "pspadm.ht_psp_state_edi_tax_file" does not exist
psql:CREATE-TRIGGER.sql:375: ERROR:  relation "pspadm.ht_psp_system_payment_requirem" does not exist
psql:CREATE-TRIGGER.sql:383: ERROR:  relation "pspadm.ht_psp_system_requirement" does not exist
psql:CREATE-TRIGGER.sql:391: ERROR:  relation "pspadm.ht_psp_tax_company_service_inf" does not exist
psql:CREATE-TRIGGER.sql:399: ERROR:  relation "pspadm.ht_psp_threshold_requirement" does not exist
psql:CREATE-TRIGGER.sql:407: ERROR:  relation "pspadm.ht_psp_tp401kcompany_service_i" does not exist
psql:CREATE-TRIGGER.sql:415: ERROR:  relation "pspadm.quest_sl_temp_explain1" does not exist
psql:CREATE-TRIGGER.sql:423: ERROR:  relation "pspadm.quest_sl_temp_explain2" does not exist
psql:CREATE-TRIGGER.sql:431: ERROR:  relation "pspadm.sys_temp_fbt" does not exist
psql:CREATE-TRIGGER.sql:431: ERROR:  function pspadm.sys_temp_fbt_iud() does not exist
```

Views

```
1. VIEW pspadm.quest_sl_temp_explain1
psql:CREATE-VIEW.sql:156: ERROR:  function pspadm.quest_sl_temp_explain1() does not exist
LINE 2: SELECT * FROM pspadm.quest_sl_temp_explain1();
                      ^
2. VIEW pspadm.quest_sl_temp_explain2
HINT:  No function matches the given name and argument types. You might need to add explicit type casts.
psql:CREATE-VIEW.sql:161: ERROR:  function pspadm.quest_sl_temp_explain2() does not exist
LINE 2: SELECT * FROM pspadm.quest_sl_temp_explain2();
                      ^
3.
HINT:  No function matches the given name and argument types. You might need to add explicit type casts.
```

Indexes

```
1.
CREATE INDEX psp_company_event_detail_i1
ON pspadm.psp_company_event_detail
USING BTREE (event_detail_type_cd ASC, value ASC, company_fk ASC);

psql:CREATE-INDEX.sql:473: ERROR:  index row size 2952 exceeds maximum 2712 for index
"psp_company_event_detail_i1"
HINT:  Values larger than 1/3 of a buffer page cannot be indexed.
Consider a function index of an MD5 hash of the value, or use full text indexing.

2.
CREATE UNIQUE INDEX psp_paycheck_split_u1
ON pspadm.psp_paycheck_split
USING BTREE (source_dd_txn_id ASC, paycheck_fk ASC);

psql:CREATE-INDEX.sql:1895: ERROR:  insufficient columns in UNIQUE constraint definition
DETAIL:  UNIQUE constraint on table "psp_paycheck_split" lacks column "created_date" which is part of the
partition key.
```

Constraints

```
psql:CREATE-CHECK-CONSTRAINT.sql:468: ERROR:  check constraint "c_psp_batch_job_status0" is violated by some row
psql:CREATE-CHECK-CONSTRAINT.sql:590: ERROR:  check constraint "c_psp_company_event_detail0" is violated by
some row
psql:CREATE-CHECK-CONSTRAINT.sql:601: ERROR:  check constraint "c_psp_company_event_email1" is violated by some
row
psql:CREATE-CHECK-CONSTRAINT.sql:1585: ERROR:  check constraint "c_psp_event_log0" is violated by some row
psql:CREATE-CHECK-CONSTRAINT.sql:3515: ERROR:  check constraint "c_psp_transaction_return_b0" is violated by
some row
```

Solution: Had to create with NOT VALID option for above constraints.

Check constraints for below tables. Need to correct

- PSP_ENTRY_DETAIL_RECORD
- PSP_FINANCIAL_TRANSACTION
- PSP_LEDGER_BALANCE
- PSP_MONEY_MOVEMENT_TRANSACTION
- PSP_PAYCHECK

Solution:

Check constraints have been fixed for all of above including partitioned tables.

Foreign key constraints

- Since we have bad data sitting in SYS database. In oracle we have NO VALIDATE constraint on tables but in APG 12 we cannot add NOT VALID (NOT VALIDATED) foreign key constraint on partitioned table referring another partitioned/non-partitioned table.
- Missing 32 FK out of which 2 were because of bad data in tables and hence could not add NOT VALID constraint on partitioned tables.

```
1.
ALTER TABLE pspadm.psp_company_agency
ADD CONSTRAINT psp_company_agency_fk2 FOREIGN KEY (company_fk, realm_id)
REFERENCES pspadm.psp_company (company_seq, realm_id)
ON DELETE NO ACTION;

psql:CREATE-FOREIGN-KEY-CONSTRAINT.sql:384: ERROR:  insert or update on table "psp_company_agency" violates
foreign key constraint "psp_company_agency_fk2"
DETAIL:  Key (company_fk, realm_id)=(dec610ad-e1ed-4846-bdb6-9b19351267ba, -1) is not present in table
"psp_company".

2.
ALTER TABLE pspadm.psp_company_bank_account
ADD CONSTRAINT psp_company_bank_account_fk1 FOREIGN KEY (bank_account_fk, realm_id)
REFERENCES pspadm.psp_bank_account (bank_account_seq, realm_id)
ON DELETE NO ACTION;

psql:CREATE-FOREIGN-KEY-CONSTRAINT.sql:391: ERROR:  insert or update on table "psp_company_bank_account"
violates foreign key constraint "psp_company_bank_account_fk1"
DETAIL:  Key (bank_account_fk, realm_id)=(6fe7d904-da36-4afe-bd91-80d886272007, -1) is not present in table
"psp_bank_account".

3.
ALTER TABLE pspadm.psp_company_service
ADD CONSTRAINT psp_company_service_fk1 FOREIGN KEY (company_fk, realm_id)
REFERENCES pspadm.psp_company (company_seq, realm_id)
ON DELETE NO ACTION;

psql:CREATE-FOREIGN-KEY-CONSTRAINT.sql:587: ERROR:  insert or update on table "psp_company_service" violates
foreign key constraint "psp_company_service_fk1"
DETAIL:  Key (company_fk, realm_id)=(dec610ad-e1ed-4846-bdb6-9b19351267ba, -1) is not present in table
"psp_company".

4.
ALTER TABLE pspadm.psp_employee_bank_account
ADD CONSTRAINT psp_employee_bank_account_fk1 FOREIGN KEY (bank_account_fk, realm_id)
REFERENCES pspadm.psp_bank_account (bank_account_seq, realm_id)
ON DELETE NO ACTION;

psql:CREATE-FOREIGN-KEY-CONSTRAINT.sql:923: ERROR:  insert or update on table "psp_employee_bank_account"
violates foreign key constraint "psp_employee_bank_account_fk1"
DETAIL:  Key (bank_account_fk, realm_id)=(792349e5-3d6a-4859-bd21-e7920831b1dd, -1) is not present in table
"psp_bank_account".

5.
ALTER TABLE pspadm.psp_liability_check
ADD CONSTRAINT psp_liability_check_fk1 FOREIGN KEY (company_fk, realm_id)
REFERENCES pspadm.psp_company (company_seq, realm_id)
ON DELETE NO ACTION;

psql:CREATE-FOREIGN-KEY-CONSTRAINT.sql:1452: ERROR:  insert or update on table "psp_liability_check" violates
foreign key constraint "psp_liability_check_fk1"
DETAIL:  Key (company_fk, realm_id)=(dec610ad-e1ed-4846-bdb6-9b19351267ba, -1) is not present in table
"psp_company".
```

DMS Validation

Documentation reference: https://docs.aws.amazon.com/dms/latest/userguide/CHAP_Validating.html

Partitioned Tables

Since partitioned table does not have global primary key, DMS does not compare with message "No Primary Key".

Tables with CLOB column

Most of tables having CLOB columns, DMS validation job reported "Mismatched records" [Failure_type: RECORD_DIFF].

	TABLE_NAME	Validation state	Remarks
1	PSP_ACHENROLLMENT_FILE	Mismatched records	OOS CLOB column: file_content, Failure type: record_diff
2	PSP_LEDGER_OPERATION_JOB	Mismatched records	OOS CLOB columns: (original_file, processed_file), Failure type: record_diff
3	PSP_STATE_REPORT_OUTPUT	Mismatched records	OOS CLOB column: report_output, Failure type: record_diff
4	PSP_PROPERTY_AUDIT	Table error	Has many OOS rows. Need to revalidate. Failure type: missing_source Manually checked rowcount are matching

Aurora PostgreSQL to Oracle replication

Limitations while using DMS (Aurora PostgreSQL to Oracle)

- Limitation on using Oracle RDS as target

https://docs.aws.amazon.com/dms/latest/userguide/CHAP_Target.Oracle.html#CHAP_Target.Oracle.Limitations

- Limitations on using PostgreSQL as a source for AWS Database Migration Service

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- Table partitioning limitations as of PostgreSQL 11 (reference: https://www.postgresql.org/docs/11/ddl-partitioning.html#DDL-PARTITIONING-DECLARATIVE)

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- `BEFORE ROW` triggers, if necessary, must be defined on individual partitions, not the partitioned table.
- Mixing temporary and permanent relations in the same partition tree is not allowed. Hence, if the partitioned table is permanent, so must be its partitions and likewise if the partitioned table is temporary. When using temporary relations, all members of the partition tree have to be from the same session.